

Online Examination Report

Date:	21.07.2014
Attempts:	1
Examinee:	Ayul, Paul
Date of birth:	11/10/1991
Subject:	040-Human Performance
Result:	60 %
Time used:	00:00:13 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	209	40.	HUMAN PERFORMANCE	0,00	1,00
2	175	40.	HUMAN PERFORMANCE	1,00	1,00
3	169	40.	HUMAN PERFORMANCE	1,00	1,00
4	193	40.	HUMAN PERFORMANCE	0,00	1,00
5	163	40.	HUMAN PERFORMANCE	0,00	1,00
6	204	40.	HUMAN PERFORMANCE	0,00	1,00
7	177	40.	HUMAN PERFORMANCE	0,00	1,00
8	195	40.	HUMAN PERFORMANCE	1,00	1,00
9	184	40.	HUMAN PERFORMANCE	1,00	1,00
10	20	40.1.2	Accident statistics	0,00	1,00
11	55	40.2.1.1	The atmosphere	0,00	1,00
12	16	40.2.1.1	The atmosphere	1,00	1,00
13	33	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
14	71	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
15	38	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
16	18	40.2.1.2	Respiratory and circulatory systems	0,00	1,00
17	34	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
18	88	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
19	47	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
20	45	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
21	19	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
22	13	40.2.2.2	Vision	0,00	1,00
23	42	40.2.2.2	Vision	0,00	1,00
24	27	40.2.2.2	Vision	0,00	1,00
25	46	40.2.2.3	Hearing	1,00	1,00
26	107	40.2.3	Health and hygiene	1,00	1,00
27	28	40.2.3.3	Problem areas for pilots	0,00	1,00
28	32	40.2.3.3	Problem areas for pilots	1,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	4	40.2.3.3	Problem areas for pilots	1,00	1,00
30	87	40.2.3.4	Intoxication	0,00	1,00
31	2	40.3	BASIC AVIATION PSYCHOLOGY	1,00	1,00
32	75	40.3.1.4	Response selection	1,00	1,00
33	21	40.3.2.4	Error generation	1,00	1,00
34	5	40.3.4.3	Co-operation	1,00	1,00
35	26	40.3.4.4	Communication	1,00	1,00
36	15	40.3.4.4	Communication	1,00	1,00
37	82	40.3.5.1	Personality, attitude and behaviour	0,00	1,00
38	92	40.3.5.1	Personality, attitude and behaviour	0,00	1,00
39	7	40.3.6.1	Arousal	1,00	1,00
40	48	40.3.6.2	Stress	0,00	1,00
Total:60%				24,00	40,00

Attachment 2: List of result by topic

Licence

Date: 18.07.2014

Examinee: Ayul, Paul

Location: Uganda Civil Aviation Authority

Your Result: 24,00 from 40,00 Points (60,00 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
40. HUMAN PERFORMANCE	40	24,00	40,00	60,00
40.1. Human Factors: basic concepts	1	0,00	1,00	0,00
40.2. Basic aviation physiology and health maintenance	20	13,00	20,00	65,00
40.3. BASIC AVIATION PSYCHOLOGY	10	7,00	10,00	70,00
Total	40	24,00	40,00	60,00

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Date of birth: 1991-11-10

Subject: 040-Human Performance
Date: 2014-07-18

1

What effect does haze have on the ability to see traffic or terrain features during flight?

- ☐ Haze creates the illusion of being a greater distance than actual from the runway, and causes pilots to fly a lower approach.
- ☐ The eyes tend to overwork in haze and do not detect relative movement easily.
- ☒ Haze causes the eyes to focus at infinity, making terrain features harder to see.

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2

Examples of classic behavioral traps that experienced pilots may fall into are to

- ☐ promote situational awareness and then necessary changes in behavior.
- ☐ assume additional responsibilities and assert PIC authority.
- ☒ complete a flight as planned, please passengers, meet schedules, and 'get the job done.'

3

Motion sickness is caused by

- ☒ continued stimulation of the tiny portion of the inner ear which controls sense of balance.
- ☐ the movement of an aircraft causing the stomach to create an acid substance which causes the stomach lining to contract.
- ☐ an instability in the brain cells which affects balance and will generally be overcome with experience.

4

Due to visual illusion, when landing on a narrower-than-usual runway, the aircraft will appear to be

- ☐ higher than actual, leading to a lower-than-normal approach.
- ☐ higher than actual, leading to a higher-than-normal approach.
- ☒ lower than actual, leading to a higher-than-normal approach.

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5

Which among these is a result of hyperventilation?

- ☒ breathing too rapidly causing a lack of oxygen.
- ☐ a need to increase the flow of supplemental oxygen.
- ☐ a lack of carbon dioxide in the body.

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6

A pilot is more subject to spatial disorientation if

- ☐ body signals are used to interpret flight attitude.
- ☒ eyes are moved often in the process of cross-checking the flight instruments.
- ☐ kinesthetic senses are ignored.

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7

In order to gain a realistic perspective on one's attitude toward flying, a pilot should

- ☒ obtain both realistic and thorough flight instruction during training.
- ☐ understand the need to complete the flight.
- ☐ take a Self-Assessment Hazardous Attitude Inventory Test.

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8

Abrupt head movement during a prolonged constant rate turn in IMC or simulated instrument conditions can cause

☒ **pilot disorientation.**

☐ **elevator illusion.**

☐ **false horizon.**

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9

What is the antidote for a pilot with a 'macho' attitude?

☐ Follow the rules. They are usually right.

☒ Taking chances is foolish.

☐ I'm not helpless. I can make a difference.

10

The distribution of primary causes of accidents in the worldwide jet aircraft commercial fleet shows that human error is involved in:

- ☐ 70% of cases
- ☒ about 90% of cases
- ☐ about 70% of cases
- ☐ all cases, one way or another

11

Air at an altitude of 18.000 feet contains, approximately:

☐ 5% oxygen

☒ 10% oxygen

☐ 21% oxygen

☐ 15% oxygen

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12

Which gas below 70,000 ft., above ground gas makes up the major part of the atmosphere ?

☐ Ozone

☒ Nitrogen

☐ Carbon dioxide

☐ Oxygen

13

Hyperventilation is:

- ☐ an increased heart rate caused by an increasing blood pressure
- ☐ a reduction of partial oxygen pressure in the brain
- ☐ an increased heart rate caused by a decreasing blood-pressure
- ☒ a normal compensatory physiological reaction to a drop in partial oxygen pressure
(i.e. when climbing a high mountain)

14

When exhaling, the expired air contains:

- ☐ less water vapour than in the inhaled air
- ☐ more nitrogen than in the inhaled air
- ☒ more carbon dioxide than in the inhaled air
- ☐ more oxygen than in the inhaled air

15

Flying a non-pressurized light aircraft at 9 000 feet your passenger develops blue lips:

- ☐ you supply additional oxygen but controlled.
- ☐ you supply him/her with extra oxygen, but do not reduce altitude .
- ☒ you reduce altitude and supply him/her with oxygen.
- ☐ calm him/her and occupy him with suitable tasks.

16

The rate and depth of breathing is primarily controlled by:

- ☒ the total atmospheric pressure
- ☐ the amount of carbon dioxide in the blood
- ☐ the amount of nitrogen in the blood
- ☐ the amount of carbon monoxide in the blood

17

Symptoms of decompression sickness

- ☐ are flatulence and pain in the middle ear
- ☒ are bends, chokes, creeps and neurological symptoms
- ☐ can only develop at altitudes of more than 40000 FT
- ☐ are only relevant when diving

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Subject: 040-Human Performance
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18

A pilot can prevent hypoxia by:

- ☐ not exceeding a cabin pressure altitude of 20000 ft
- ☒ using additional oxygen when flying above 10000 ft
- ☐ swallowing, yawing and applying the Valsalva method
- ☐ relying on the body's built in warning system recognizing any stage of hypoxia

19

The respiratory process consists mainly of

- ☐ the transportation of carbon dioxide to the cells and elimination of oxygen
- ☒ the diffusion of oxygen through the respiratory membranes into the blood, transportation to the cells, diffusion into the cells and elimination of carbon dioxide from the body
- ☐ the transportation of oxygen to the cells and the elimination of nitrogen
- ☐ the transportation of oxygen to the cells and the elimination of carbon monoxide

20

The total gas volume of the lung is the sum of:

1. tidal volume
2. inspiratory reserve volume
3. expiratory reserve volume
4. residual volume

Which of the following lists the correct combination?

- ☐ 1 and 2
- ☐ 1, 2 and 3
- ☐ 2 and 3
- ☒ 1, 2, 3 and 4

21

Hypoxia is:

- ☐ a condition of lacking oxygen in the brain causing the circulatory system to compensate by decreasing the heart rate.
- ☐ a physical condition caused by a lack of oxygen saturation in the blood while hyperventilating.
- ☒ a physical condition caused by a lack of oxygen to meet the needs of the body tissues, leading to mental and muscular disturbances
- ☐ often produced during steep turns when pilots turn their heads in a direction opposite to the direction in which the aircraft is turning

22

When focussing on near objects:

- ☐ the shape of lens gets flatter
- ☐ the pupil gets larger
- ☒ the shape of lens gets more spherical
- ☒ the cornea gets smaller

23

When flying through a thunderstorm with lightning you can protect yourself from flashblindness by:

- 1 - turning up the intensity of cockpit lights**
- 2 - looking inside the cockpit**
- 3 - wearing sunglasses**
- 4 - using blinds or curtains when installed**

☐ **1 and 2 are correct, 3 and 4 are false**

☐ **1, 2 and 3 are correct, 4 is false**

☐ **1, 2, 3 and 4 are correct**

☒ **3 and 4 are correct, 1 and 2 are false**

24

Illuminated anti-collision lights in IMC

- ☒ will improve the pilots depth perception
- ☐ can cause colour-illusions
- ☐ can cause disorientation
- ☐ will effect the pilots binocular vision

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25

Any prolonged exposure to noise in excess of 90 db can result in:

- ☒ noise induced hearing loss
- ☐ presbycusis (effects of aging)
- ☐ a ruptured ear drum
- ☐ conductive hearing loss

26

Which of these is true regarding the presence of alcohol within the human body?

- ☐ An increase in altitude decreases the adverse effect of alcohol.
- ☒ Judgment and decision-making abilities can be adversely affected by even small amounts of alcohol.
- ☐ A small amount of alcohol increases vision acuity.

27

Which of the following statements about hyperthermia is correct ?

- ☒ Vasodilatation is the only mechanism which is capable of reducing body temperature.
- ☐ Performance is not impaired by an increase in body temperature to 40°C or more.
- ☐ Evaporation is more effective when ambient humidity is high.
- ☐ Complete adaptation to the heat in a hot country takes about a fortnight.

28

Flying with a "common cold":

- ☐ increases the risk of hypertension.
- ☒ may lead to incapacitation due to severe sinus or ear pain.
- ☐ will cause infection in other crew members if you are flying in a pressurised aircraft.
- ☐ is permitted as long as you are on treatment with antibiotics.

29

How is Yellow fever contracted?

☒ **A virus transmitted by an infected mosquito.**

☐ **Contaminated food or water.**

☐ **By contact with the saliva of infected animals.**

☐ **Excessive use of alcohol.**

30

Which statement is correct?

1. Smokers have a greater chance of suffering from coronary heart disease
2. Smoking tobacco will raise the individuals physiological altitude during flight
3. Smokers have a greater chance of contracting lung cancer

- ☐ 1,2 and 3 are correct
- ☒ 1 and 3 are correct, 2 is false
- ☐ 2 and 3 are correct, 1 is false
- ☐ 1 and 2 are correct, 3 is false

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31

What are two types of attention ?

- ☒ Selective and divided
- ☐ Divided and behavioural
- ☐ Cognitive and intuitive
- ☐ Intuitive and behavioural

32

Planning:

- ☐ is dangerous in the cockpit, as it interrupts flight crew creativity
- ☒ allows crew members to anticipate potential risky situations and decide on possible responses
- ☐ in the cockpit typically results in plans that are always easy to modify when things are not as anticipated
- ☐ is unnecessary in the cockpit, as crew members are so highly trained, they will always know what to do in unusual situations

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33

Human error rates during the performance of a simple and repetitive task can normally be expected to be approximately:

☒ 1 in 100

☐ 1 in 500

☐ 1 in 2000

☐ 1 in 200

34

Which of the following statements best characterises a self-centered cockpit ?

- ☐ The egoistic and self-centered personality of the captain often leads to a synergetic cockpit
- ☐ While decreasing communication, the independence of each member bolsters the crew's synergy
- ☐ The communication between crew members always increases when the captain takes charge of a situation
- ☒ Each one does his own thing while at the same time assuming that everyone is aware of what is being done or what is going on

35

Which of the following are strategies for resolving conflict?

1. Seeking arbitration
2. Actively listening to other people
3. Recognise the early signs of conflict and address them
4. Becoming aware of cultural influences

☒ 1, 2, 3 and 4

☐ 2, 3, 4 only

☐ 1, 2, 4 only

☐ 1, 2, 3 only

36

With regard to the practice of English, which of the following statements is correct?

- ☐ Be familiar with normal procedures in English since only this allows for effective management of any flight's communication.
- ☐ The composition of every crew should be geared to a command of the official aeronautical language of the destination country.
- ☐ It is necessary and sufficient to have a command of any of the official languages of the ICAO.
- ☒ All pilots should have a sound understanding of English because it is safer to adhere to the common language chosen by the aeronautical world.

37

Personality is based on:

1. Heredity
2. Upbringing
3. Experience
4. Childhood

☐ None of the listed answers is correct

☐ 1, 2 and 4 only are correct

☐ 1, 2, 3 and 4 are correct

☒ 2, 3 and 4 only are correct

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38

Which of the following elements make up the personality of an individual?

1. Heredity
2. Childhood environment
3. Upbringing
4. Past experience

☐ 1,2,3,4

☐ 2,3

☒ 2,3,4

☐ 1,2,4

39

An identical situation can be experienced by one pilot as exciting in a positive sense and by another pilot as threatening. In both cases:

- ☒ **the arousal level of both pilots will be raised**
- ☐ **both pilots will experience the same amount of stress**
- ☐ **the pilot feeling threatened, will be much more relaxed, than the pilot looking forward to what may happen**
- ☐ **both pilots will loose their motor-coordination**

40

The behavioural effects of stress may include:

- 1: manifestation of aggressiveness.
- 2: a willingness to improve communication.
- 3: a willingness for group cohesion.
- 4: a tendency to withdrawal.

The combination of correct statements is:

- ☐ 3 and 4 are correct
- ☐ 2,3 and 4 are correct
- ☒ 1,2 and 3 are correct
- ☐ 1 and 4 are correct